THE TRANSITIVITY DISTINGUISHED TRIANGLE OF NAIVE COTANGENT COMPLEXES FOR A COMPLETE INTERSECTION MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3933-3943 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3945-3958, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The canonical map and six-term sequence of Modules, Lemma exact-sequence-NL-ringed-topoi, the affine description of naive cotangent complexes (lemma-NL-affine, same chapter), and the algebra transitivity result for lci morphisms (More on Algebra); cited inputs with the reduction written in the retained span.

Source reference. Original lemma 0D0I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3736–3745. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E44: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3916–3926. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E1Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3958–3958. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07D4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3958–3958. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ and $Y \to Z$ be morphisms of schemes. Assume $X \to Y$ is a complete intersection morphism. Then there is a canonical distinguished triangle*

$$f^* NL_{Y/Z} \to NL_{X/Z} \to NL_{X/Y} \to f^* NL_{Y/Z}[1]$$

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in $D(\mathcal{O}_X)$ which recovers the 6-term exact sequence of Lemma 0E44.

Proof. It suffices to show the canonical map

$$f^* NL_{Y/Z} \to \text{Cone}(NL_{X/Z} \to NL_{X/Y})[-1]$$

of Modules, Lemma 0E1Z is an isomorphism in $D(\mathcal{O}_X)$. In order to show this, it suffices to show that the 6-term sequence has a zero on the left, i.e., that $H^{-1}(f^* NL_{Y/Z}) \to H^{-1}(NL_{X/Z})$ is injective. Affine locally this follows from the corresponding algebra result in More on Algebra, Lemma 07D4. To translate into algebra use Lemma 0D0I. □